Society for Computer Technology & Research's

PUNE INSTITUTE OF COMPUTER TECHNOLOGY, Pune.

DEPARTMENT OF INFORMATION TECHNOLOGY

MINI PROJECT USING DATA STUCUTURE AND APPLICATIONS

Class: SY IT **Division:** 09 **Group No.:** 2

Group members:

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LogiTrack: Smart Logistics and Parcel Management System

1. **Problem statement:**
   Courier distribution hubs handle hundreds of incoming parcels daily across multiple zones, facing operational bottlenecks: slow parcel tracking, lack of delivery urgency prioritization, inability to undo accidental dispatches, and unorganized loading of parcel. LogiTrack resolves these challenges by automating hub operations using core data structures for instant parcel indexing, priority-driven van dispatch, dynamic LIFO error recovery, and sorted manifest generation.

2. **Objectives:**
   - To implement an open-addressed Hash Table with circular Linear Probing for near O(1) parcel status tracking, lookups, and persistent CSV storage.
   - To implement a Priority Queue for urgency-based parcel dispatch Express, Standard & Economy and a dynamic Singly Linked List Stack enabling instant LIFO undo of accidental dispatches.

3. **Data structures used (Data structure(s) and 1–2 line justification for each):**
   - **Hash Table with Linear Probing (member1_hash.h):** Uses the modulo division hash function (id % 20) with circular linear probing for collision resolution, providing O(1) average time complexity for rapid tracking lookups, status updates, and state persistence via shipments.csv.

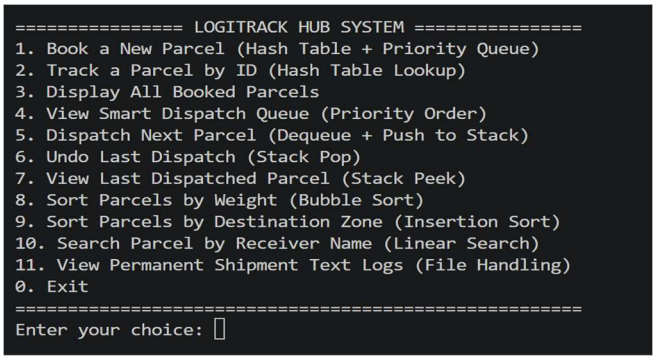
- **Priority Queue (member2_pq.h):** Uses a sorted array where incoming parcels are inserted based on urgency tier (Priority 3 = Express, 2 = Standard, 1 = Economy), ensuring high-priority shipments are always dispatched first (O(1) dequeue).

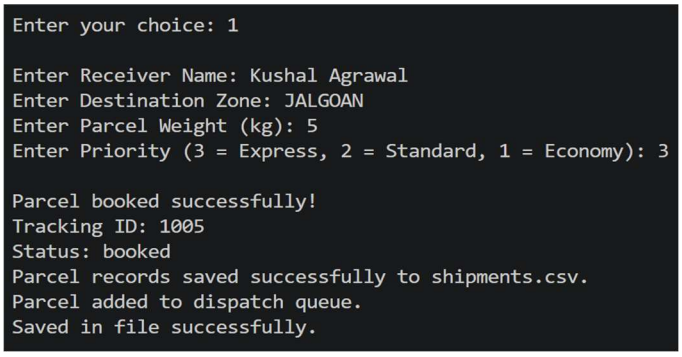
- **Dynamic Singly Linked List Stack (member3_stack.h):** Employs a dynamic LIFO stack (Node* top) to store dispatched shipments, enabling hub operators to instantly undo (pop) accidental dispatches in O(1) time without fixed-size array limits.

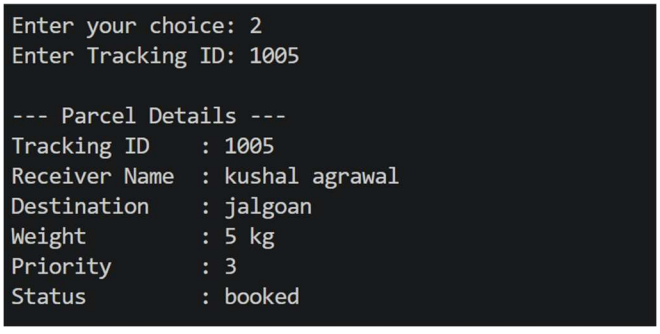
- **Bubble Sort, Insertion Sort & Linear Search (member4_sort.h):** Uses Bubble Sort to arrange manifests by parcel weight for van load verification, Insertion Sort to group parcels alphabetically by destination zone for route planning, and Linear Search to find parcels by receiver name in O(N).

4. **Implementation (Programming language / tools):** C++ (Modular structure with 4 header files and main.cpp), MinGW/GCC Compiler, Visual Studio Code (VS Code), CSV and Text File Handling.

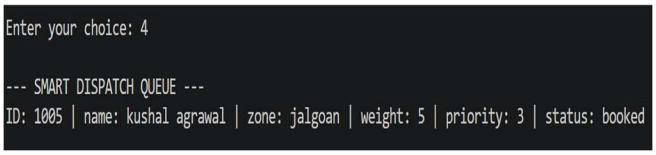
5. **Implemented output:**

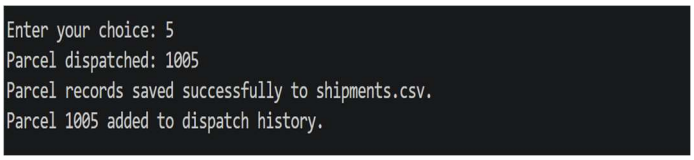












```
Enter your choice: 6
Undo successful for parcel ID: 1005
Parcel records saved successfully to shipments.csv.
Parcel added to dispatch queue.
```

```
Enter your choice: 7
Last dispatched parcel details:
ID: 1005  Name: kushal agrawal  Zone: jalgoan
```

```
Enter your choice: 8

Parcels sorted by weight:

================================================
              PARCEL DETAILS
================================================
Parcel ID   Receiver              Zone        Weight
------------------------------------------------
1002        yash                  latur       4
1001        bharat chitlange      pune        5
1005        kushal agrawal        jalgoan     5
1003        akanksha dhoot        yavatmal    7
1004        dhiren lodha          pcmc        9
================================================
```

```
Enter your choice: 9

Parcels sorted by destination zone:

================================================
              PARCEL DETAILS
================================================
Parcel ID   Receiver              Zone        Weight
------------------------------------------------
1005        kushal agrawal        jalgoan     5
1002        yash                  latur       4
1004        dhiren lodha          pcmc        9
1006        aaryan                pcmc        4
1001        bharat chitlange      pune        5
1003        akanksha dhoot        yavatmal    7
================================================
```

```
Enter your choice: 10
Enter receiver name: Bharat chitlange

Parcel found!
Tracking ID: 1001
Receiver: bharat chitlange
Zone: pune
Weight: 5
```

6. **Conclusion:**

   Successfully developed an integrated logistics hub system demonstrating how combining Hash Tables, Priority Queues, Stacks, and Sorting algorithms drastically optimizes parcel lookups, enforces critical delivery schedules, and guarantees error resilience. The project provided practical insights into data structure modularity, time-space trade-offs, collision handling, and dynamic memory management.

Student signatures:

1. Bharat Chitlange: _____
2. Yash Brahmecha: _____
3. Akanksha Dhoot: _____
4. Dhiren Lodha: _____

Faculty remarks: